Chapter 6
Multiple Integrals

Multiple Integration: The integrals of more than one variable are known as **multiple integrals** and are evaluated by a process involving iterated integrals.

Partial Integration: The process in which the integration is performed with respect to one variable treating the other variable(s) as constant, is called **partial integration**.

Iterated Integral: A definite integral which is evaluated stage by stage using partial integration is called an iterated (successive or repeated) integral.

Double Integrals: The double integral may be defined geometrically in much the same way as the definite Riemann integral. Suppose $f(x,y)$ is continuous and single-valued function of x and y both inside and boundary of R in $\mathbb{R}^2$.

Properties of Double Integrals:

1. $\iint_R cf(x,y)dA = c\iint_R f(x,y)\,dA$ where c is a constant
2. $\iint_R [f(x,y)+g(x,y)]dA = \iint_R f(x,y)\,dA + \iint_R g(x,y)dA$
3. $\iint_R [f(x,y)-\boldsymbol{g}(x,y)]dA = \iint_R f(x,y)\,dA - \iint_R g(x,y)dA$

Example 1: Evaluate the iterated integral $\int_0^1\int_1^2 (x^2+xy)dxdy$.

Solution: $$\int_0^1\int_1^2 (x^2+xy)\,dxdy = \int_0^1\left[\int_1^2 (x^2+xy)\,dx\right]dy = \int_0^1\left[\frac{x^3}{3}+\frac{x^2y}{2}\right]_{x=1}^{x=2}dy = \int_0^1\left[\frac{8-1}{3}+\frac{(4-1)y}{2}\right]dy$$

$$= \int_0^1\left[\frac{7}{3}+\frac{3y}{2}\right]dy = \left[\frac{7}{3}y+\frac{3}{2}\frac{y^2}{2}\right]_{y=0}^{y=1} = \frac{37}{12}.$$

Example 2: Evaluate the double integral $\iint_R y^2x\,dA$ over the rectangle.

$$R = \{(x,y): -3 \le x \le 2, 0 \le y \le 1\}$$

Solution: $$\iint_R y^2x\,dA = \int_0^1\int_{-3}^2 y^2x\,dx\,dy = \int_0^1 y^2\left[\int_{-3}^2 x\,dx\right]dy = \int_0^1 y^2\left[\frac{x^2}{2}\right]_{-3}^{2}$$

$$= \int_0^1 -\frac{5}{2}y^2\,dy = -\frac{5}{2}\left[\frac{y^3}{3}\right]_{y=0}^{y=1} = -\frac{5}{2}\times\frac{1}{3} = -\frac{5}{6}.$$

Example 3: Use a double integral to find the volume of the solid that is bounded above by the plane $z = 4-x-y$ and below by the rectangle $R = [0,1]\times[0,2]$.

Solution: The volume is double integral of $z = 4-x-y$ over R.

$$V = \iint_R (4-x-y)\,dA = \int_0^2\int_0^1 (4-x-y)\,dx\,dy = \int_0^2\left[\int_0^1 (4-x-y)\,dx\right]dy$$

$$= \int_0^2\left[4x-\frac{x^2}{2}-xy\right]_{x=0}^{x=1}dy = \int_0^2\left[\frac{7}{2}-y\right]dy = \left[\frac{7}{2}y-\frac{y^2}{2}\right]_{y=0}^{2} = 5$$

Exercise Set 6.1

Evaluate the iterated integral:

1. $\int_{1}^{4}\int_{0}^{2}(6x^2y-2x)\,dy\,dx$
2. $\int_{0}^{1}\int_{0}^{1}(x+y)^2\,dx\,dy$
3. $\int_{0}^{1}\int_{1}^{2}(x+e^{-y})\,dx\,dy$
4. $\int_{0}^{\pi/6}\int_{0}^{\pi/2}(\sin x+\sin y)\,dy\,dx$
5. $\int_{-3}^{3}\int_{0}^{\pi/2}(y+y^2\cos x)\,dx\,dy$
6. $\int_{1}^{3}\int_{1}^{5}\frac{\ln y}{xy}\,dy\,dx$
7. $\int_{1}^{4}\int_{1}^{2}\left(\frac{x}{y}+\frac{y}{x}\right)dy\,dx$
8. $\int_{0}^{1}\int_{0}^{2}y\,e^{x-y}\,dx\,dy$

[1-8] (Ref: Calculus- James Stewart- 8th Ed. Page: 999-1000)

9. $\int_{0}^{1}\int_{0}^{2}(x+3)\,dy\,dx$
10. $\int_{1}^{3}\int_{-1}^{1}(2x-4y)\;dy\,dx$
11. $\int_{2}^{4}\int_{0}^{1}x^2y\;dx\;dy$
12. $\int_{-2}^{0}\int_{-1}^{2}(x^2+y^2)\,dx\;dy$
13. $\int_{0}^{\ln 3}\int_{0}^{\ln 2}e^{x+y}\,dy\,dx$
14. $\int_{0}^{2}\int_{0}^{1}y\sin x\;dy\,dx$
15. $\int_{-1}^{0}\int_{2}^{5}dx\,dy$
16. $\int_{4}^{6}\int_{-3}^{7}dy\;dx$
17. $\int_{4}^{6}\int_{-3}^{7}dy\;dx$
18. $\int_{\pi/2}^{\pi}\int_{1}^{2}x\cos x\,y\,dy\;dx$

[**9-18**] (Ref: Calculus- Howard Anton- 10th Ed. Page: 1007)

Exercise Set 6.2

Evaluate the double integral over the rectangular region *R*.

1. $\iint_R x\,sec^2\,y\,dA, R = \left\{(x,y)|0 \le x \le 2, 0 \le y \le \frac{\pi}{4}\right\}$

2. $\iint_R (y + xy^{-2})dA, R = \{(x,y)|0 \le x \le 2, 1 \le y \le 2\}$

3. $\iint_R \frac{xy^2}{x^2+1}\,dA, R = \{(x,y)|0 \le x \le 1, -3 \le y \le 3\}$

4. $\iint_R ye^{-xy}dA, R = [0,2] \times [0,3]$

[1-4] (Ref: Calculus- James Stewart- 8th Ed. Page:1000)

5. $\iint_R 4xy^3 dA; R = \{(x,y)| -1 \le x \le 1, -2 \le y \le 2\}$

6. $\iint_R x\sqrt{1-x^2}dA, R = \{(x,y)|0 \le x \le 1, 2 \le y \le 3\}$

7. $\iint_R (x\,sin\,y - y\,sin\,x)dA; R = \left\{(x,y)|0 \le x \le \frac{\pi}{2}, 0 \le y \le \frac{\pi}{3}\right\}$

[5-7] (Ref: Calculus- Howard Anton- 10th Ed. Page: 1007)

Double Integrals over the nonrectangular region:

(a) If R is the region defined by $R = \{(x,y)|a \le x \le b, f_1(x) \le y \le f_2(x)\}$, then

$$\iint_R f(x,y)dA = \int_a^b \left[\int_{f_1(x)}^{f_2(x)} f(x,y)dy\right] dx.$$

(b) If R is the region defined by $R = \{(x,y)|c \le y \le d, g_1(y) \le x \le g_2(y)\}$, then

$$\iint_R f(x,y)dA = \int_c^d \left[\int_{g_1(y)}^{g_2(y)} f(x,y)dx\right] dy.$$

Example 4: Evaluate

(a) $\int_0^1 \int_{-x}^{x^2} y^2 x dy dx$

(b) $\int_0^{\pi/3} \int_0^{\cos y} x \sin y \, dx \, dy$

Solution (a): $\int_0^1 \int_{-x}^{x^2} y^2 x dy dx =$

$$\int_0^1 \left[\int_{-x}^{x^2} y^2 x \, dy\right] dx = \int_0^1 \left[\frac{y^3 x}{3}\right]_{y=-x}^{y=x^2} dx$$

$$= \int_0^1 \left[\frac{x^7}{3} + \frac{x^4}{3}\right] dx = \left(\frac{x^8}{24} + \frac{x^5}{15}\right)\Big]_0^1 = \frac{13}{120}$$

Solution (b):

$$\int_0^{\pi/3} \int_0^{\cos y} x \sin y \, dx \, dy = \int_0^{\pi/3} \left[\int_0^{\cos y} x \sin y \, dx\right] dy$$

$$= \int_0^{\pi/3} \left[\frac{x^2}{2} \sin y\right]_{y=0}^{y=\cos y} dy = \int_0^{\pi/3} \left[\frac{1}{2} \cos^2 y \sin y\right] dy$$

$$= \left[-\frac{1}{6} \cos^3 y\right]_0^{\pi/3} = \frac{7}{48}$$

Example 5: Evaluate $\iint_D (x+2y)dA$, where D is the region bounded by the parabolas $y = 2x^2$, $y = 1 + x^2$.

Solution: The parabolas intersect when $2x^2 = 1 + x^2$, that is $x^2 = 1$, so $x = \pm 1$.

$$D = \left\{(x,y)| -1 \le x \le 1, 2x^2 \le y \le 1 + x^2\right\}$$

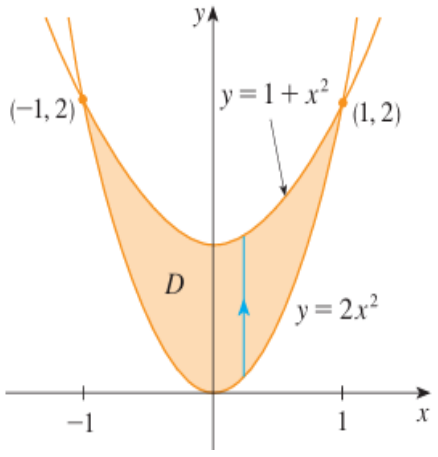
$$\iint_D (x+2y)dA = \int_{-1}^{1} \int_{2x^2}^{1+x^2} (x+2y)dy\,dx$$

$$= \int_{-1}^{1} \left[xy + y^2\right]_{y=2x^2}^{y=1+x^2} dx$$

$$= \int_{-1}^{1} \left[x\left(1+x^2\right) + \left(1+x^2\right)^2 - x\left(2x^2\right) - \left(2x^2\right)^2\right] dx$$

$$= \int_{-1}^{1} \left[-3x^4 - x^3 + 2x^2 + x + 1\right] dx$$

$$= \left[-3\frac{x^5}{5} - \frac{x^4}{4} + 2\frac{x^3}{3} + \frac{x^2}{2} + x\right]_{-1}^{1} = \frac{32}{15}$$



Exercise Set 6.3

Evaluate the iterated integral

1. $\int_1^5 \int_0^x (8x - 2y)\, dy\, dx$

2. $\int_0^2 \int_0^{y^2} x^2\, y\, dx\, dy$

3. $\int_0^1 \int_0^y x e^{y^3}\, dx\, dy$

4. $\int_0^{\pi/2} \int_0^x x \sin y\, dy\, dx$

5. $\int_0^1 \int_0^{s^2} \cos(s^3)\, dt\, ds$

[1-5] (Ref: Calculus- James Stewart- 8th Ed. Page: 1008)

6. $\int_1^5 \int_{x^2}^x x\, y^2\, dy\, dx$

7. $\int_1^{3/2} \int_y^{3-y} y\, dx\, dy$

8. $\int_0^3 \int_0^{\sqrt{9-y^2}} y\, dx\, dy$

9. $\int_{-1}^1 \int_{-x^2}^{x^2} \left(x^2 - y\right) dy\, dx$.

[6- 9] (Ref: Calculus- Howard Anton- 10th Ed. Page; 1015)

10. $\iint_D \frac{y}{x^2+1} dA;\ D = \left\{(x,y) \mid 0 \le x \le 4,\ 0 \le y \le \sqrt{x}\right\}$

11. $\iint_D (2x + y)\, dA;\ D = \left\{(x,y) \mid 1 \le y \le 2,\ y - 1 \le x \le 1\right\}$

12. $\iint_D x \cos y\, dA$; D is bounded by $y = 0,\ y = x^2,\ x = 1$

13. $\iint_D \left(x^2 + 2y\right) dA$; D is bounded by $y = x,\ y = x^3,\ x \ge 0$.

[10-13] (Ref: Calculus – James Stewart- 8th Ed. Page:1008)

14. $\iint_R x^2\, dA$, R is the region bounded by $y = \frac{16}{x},\ y = x,\ x = 8$

15. $\iint_R x\, y^2\, dA$, R is the region enclosed by $y = 1,\ y = 2,\ x = 0,\ y = x$

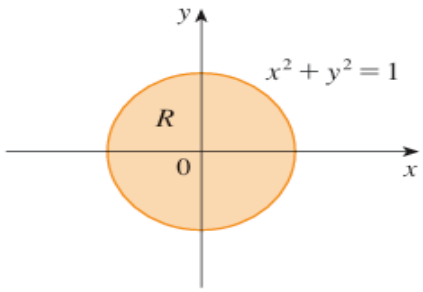
16. $\iint_R x\, y\, dA$, R is the region enclosed by $y = \sqrt{x},\ y = 6 - x,\ y = 0$

17. $\iint_R x\, y^2\, dA$, R is the region in the first quadrant enclosed between $y = x$ and $y = x^3$.

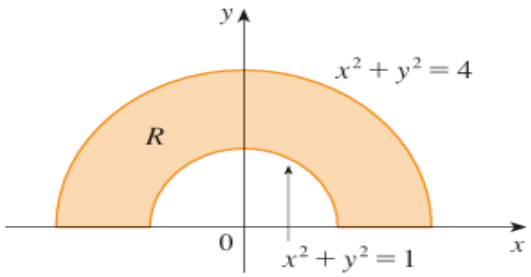
[14-17] (Ref: Calculus – Howard Anton- 10th Ed. Page # 10016)

Double Integration in polar coordinate

Suppose that we want to evaluate a double integral $\iint_R f(x,y)\,dA$, where R is one of the regions shown in Figure. In either case the description of R in terms of rectangular coordinates is rather complicated, but R is easily described using polar coordinates.



(a) $R = \{(r, \theta) \mid 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}$



(b) $R = \{(r, \theta) \mid 1 \leq r \leq 2, 0 \leq \theta \leq \pi\}$

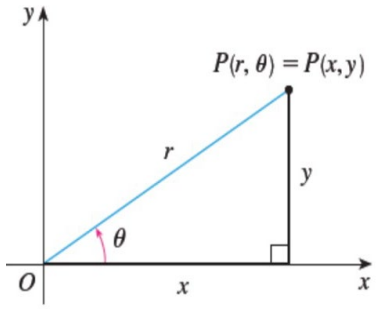
The polar coordinates (r, θ) of a point are related to the rectangular coordinate (x, y) by the equations

Change to Polar Coordinates in a Double Integral

If f is continuous in a polar rectangle R given by $a \leq r \leq b, \alpha \leq \theta \leq \beta$ Then

$$\iint_R f(x,y)\,dA = \int_{\alpha}^{\beta}\int_{a}^{b} f(r\cos\theta, r\sin\theta)\, r\,dr\,d\theta$$

$$r^2 = x^2 + y^2$$
$$x = r\cos\theta$$
$$y = r\sin\theta$$



Example 6: Evaluate $\iint_R e^{-\left(x^2+y^2\right)}\,dA$ where R is the annulus $x^2 + y^2 \geq 1, x^2 + y^2 \leq 4$.

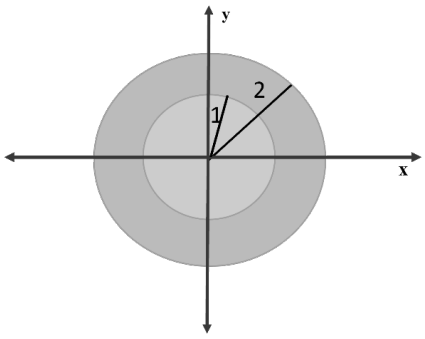
Solution:

$$\iint_R e^{-(x^2+y^2)}dA$$

$$= \int_{\theta=0}^{\theta=2\pi}\int_{r=1}^{r=2} e^{-r^2} r\,dr\,d\theta$$

$$= -\int_0^{2\pi} (e^{-4} - e^{-1})\,d\theta$$

$$= 2\pi(e^{-1} - e^{-4})$$



Exercise Set 6.4

Evaluate the given integral by changing to polar coordinates.

1. $\iint_D x^2 y \, dA$, where D is the top half of the disk with center the origin and radius 5.
2. $\iint_R (2x - y) \, dA$, where R is the region in the first quadrant enclosed by the circle $x^2 + y^2 = 4$ and the lines $x = 0$ and $y = x$.
3. $\iint_R \sin(x^2 + y^2) \, dA$, where R is the region in the first quadrant between the circles with the center of the origin and radii 1 and 3.
4. $\iint_R \frac{y^2}{x^2 + y^2} \, dA$, where R is the region that lies between the circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$ with $0 \le a \le b$.
5. $\iint_D e^{-x^2 - y^2} \, dA$, where D is the region bounded by the semi-circle $x = \sqrt{4 - y^2}$ and the $y-$axis.
6. $\iint_D \cos\sqrt{x^2 + y^2} \, dA$, where D is the disk with the center the origin and radius 2.

[1-6] (Ref: Calculus-James Stewart- 8th Ed. Page: 1014)

7. $\iint_R \sin(x^2 + y^2) \, dA$, where R is the region enclosed by the circle $x^2 + y^2 = 9$.
8. $\iint_R \sqrt{9 - x^2 - y^2} \, dA$, where R is the region in the first quadrant within the circle $x^2 + y^2 = 9$.
9. $\iint_R \frac{1}{1 + x^2 + y^2} dA$, where R is the sector in the first quadrant bounded by $y = 0$, $y = x$ and $x^2 + y^2 = 4$.

[7-9] (Ref: Calculus-Howard Anton- 10th Ed. Page: 1025)

Change of variables in a double integral

Suppose that f is continuous on R

$$\iint_R f(x,y)\,dA = \iint f\left(x(u,v), y(u,v)\right) \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv$$

Where, the Jacobian of the transformation T given by $x = g(u,v)$ and $y = h(u,v)$ is

$$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}.$$

Example 7: Evaluate the integral $\iint_R e^{(x+y)/(x-y)}\,dA$, where R is the trapezoidal region with vertices $(1,0)$, $(2,0)$, $(0,-2)$ and $(0,-1)$.

Solution: Since it isn't easy to change to integrate $e^{(x+y)/(x-y)}$, we make a change of variables suggested by the form of the function: $u = x + y, v = x - y$

These equations define a transformation T^{-1} from the xy-plane to the uw −plane. By solveing above equation for x and y

$$x = \frac{1}{2}(u+v), \quad y = \frac{1}{2}(u-v)$$

The Jacobian of T is $\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} = -\frac{1}{2}$

To find the region S in the uv-plane corresponding to R, we note that the sides of R lie on the lines

$$y = 0 \quad x - y = 2 \quad x = 0 \quad x - y = 1$$

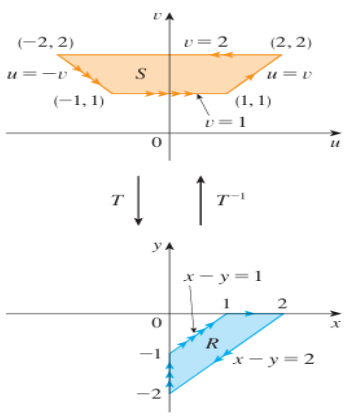
The image lines in the uv −plane are $u = v$ $v = 2$ $u = -v$ $u = 1$

Thus, region S is the trapezoidal region with vertices (1, 1), (2, 2), (-2, 2) and (-1, $S = \{(u,v) | 1 \le v \le 2, -v \le u \le v\}$

$$\iint_R e^{(x+y)/(x-y)}\,dA = \iint_S e^{u/v} \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv$$

$$= \int_1^2 \int_{-v}^{v} e^{u/v} \left(\frac{1}{2}\right) du\,dv = \frac{1}{2}\int_1^2 \left[v e^{u/v} \right]_{u=-v}^{u=v} dv$$

$$= \frac{1}{2}\int_1^2 \left(e - e^{-1}\right) v\ dv = \frac{3}{4}\left(e - e^{-1}\right).$$



Exercise Set 6.5

Evaluate the integral by making an appropriate change of variables.

1. $\iint_R \frac{x-2y}{3x-y}\,dA$, where R is the parallelogram enclosed by the lines $x-2y=0,\ x-2y=4,\ 3x-y=1$ and $3x-y=8$

2. $\iint_R (x+y)\,e^{x^2-y^2}\,dA$, where R is the rectangle enclosed by the lines $x-y=0, x-y=2,\ x+y=0$ and $x+y=3$.

[1-2] (Ref: Calculus-James Stewart- 8th Ed. Page:1060)

3. $\iint_R \frac{y-4x}{y+4x}\,dA$, where R is the region enclosed by the lines $y=4x,\ y=4x+2, y=2-4x,\ y=5-4x$.

4. $\iint_R (x^2-y^2)\,dA$, where R is the rectangular enclosed by the lines $y=-x,\ y=1-x, y=x, y=x+2$.

[3-4] (Ref: Calculus-Howard Anton- 10th Edition. Page: 1070)

Chapter 7
Application of Double Integral

Mass and center of mass: The co-ordinates $(\bar{x}, \bar{y})$of the center of mass of lamina occupying region D and having density function $\rho(x, y)$ are

$$\bar{x} = \frac{1}{m} \iint_D x\,\rho\,(x,y)\,dA \quad \text{and} \quad \bar{y} = \frac{1}{m} \iint_D y\,\rho\,(x,y)\,dA$$

Where the mass m is given by $m = \iint_D \rho\,(x,y)\,dA$

Example 1: Find the mass and center of mass of the lamina that occupies the region D and has the given density function ρ. Where $D = \{(x,y) | 0 \le x \le 1,\ 0 \le y \le 2\}$ and $\rho(x,y) = x + y$.

Solution:

$$m = \int_0^2 \int_0^1 (x+y)dx\,dy$$

$$= \int_0^2 \left(\frac{x^2}{2} + xy\right)\Big|_0^1 dy$$

$$= \int_0^2 \left(\frac{1}{2} + y\right) dy$$

$$= \left(\frac{1}{2}y + \frac{y^2}{2}\right)\Big|_0^2 = 3$$

$\bar{x} = \frac{1}{m} \iint_D x\,\rho(x,\ y)\,dA$

$\bar{x} = \frac{1}{3}\int_0^2 \int_0^1 x(x+y)dx\,dy$

$= \frac{1}{3}\int_0^2 \int_0^1 (x^2 + xy)dx\,dy$

$= \frac{1}{3}\int_0^2 \left(\frac{x^3}{3} + \frac{x^2}{2}y\right)\Big|_0^1 dy$

$= \frac{1}{3}\int_0^2 \left(\frac{1}{3} + \frac{1}{2}y\right) dy = \frac{1}{3}\left(\frac{1}{3}y + \frac{1}{2}\frac{y^2}{2}\right)\Big|_0^2$

$= \frac{5}{9}$

Therefore, Mass, $m = 3$

$\bar{y} = \frac{1}{m} \iint_D y\,\rho(x,\ y)\,dA$

$= \frac{1}{3}\int_0^2 \int_0^1 y\,(x+y)dx\,dy$

$= \frac{1}{3}\int_0^2 \int_0^1 (xy + y^2)dx\,dy$

$= \frac{1}{3}\int_0^2 \left(\frac{x^2}{2}y + xy^2\right)\Big|_0^1 dy$

$= \frac{1}{3}\int_0^2 \left(\frac{1}{2}y + y^2\right) dy$

$= \frac{1}{3}\left(\frac{1}{2}\frac{y^2}{2} + \frac{y^3}{3}\right)\Big|_0^2 = \frac{11}{9}$

and center of mass $(\bar{x}, \bar{y}) = \left(\frac{5}{9}, \frac{11}{9}\right)$

Exercise Set 7

Find the mass and center of mass of the lamina.

1. A lamina that occupies region *D* and has the given density function ρ.

   Where $D = \{(x,y)|\ 1 \le x \le 3,\ 1 \le y \le 4\}$ and $\rho(x,y) = ky^2$.

2. A lamina that occupies region *D* and has the given density function ρ.

   Where $D = \{(x,y)|\ 0 \le x \le a,\ 0 \le y \le b\}$ and $\rho(x,y) = 1 + x^2 + y^2$.

3. A lamina that occupies region *D* and has the given density function ρ.

   Where D is the triangular region with vertices $(0,0), (2,1), (0,3)$ and . $\rho(x,y) = x + y$.

4. A lamina that occupies region *D* and has the given density function ρ.

   Where D is the triangular region enclosed by the lines $y = 0, y = 2x$ and $x + 2y = 1$; $\rho(x,y) = x$.

5. A lamina that occupies region *D* and has the given density function ρ. Where D is bounded by $y = 1 - x^2,\ y = 0$ and $\rho(x,y) = ky$.

   [1-5] (Ref: Calculus-James Stewart- 8th Ed. Page: 1024- 1025)

6. A lamina that occupies region *D* and has the given density function ρ. Where D is bounded by $y = x + 2,\ y = x^2$ and $\rho(x,y) = kx^2$.

7. A lamina that occupies region *D* and has the given density function ρ. Where D is bounded by the $x-$ axis, the line $x = 1$ and the curve $y = \sqrt{x}$ and $\rho(x,y) = x + y$.

   [6-7] (Ref: Calculus-Howard Anton- 10th Edition. Page: 1077)

Chapter 8

Triple Integrals

Iterated integrals in three variables.

Example 1: Evaluate $\int_0^3 \int_0^1 \int_{-1}^1 (x^2 + yz)dzdydx$.

Solution: $\int_0^3 \int_0^1 \int_{-1}^1 (x^2 + yz)dzdydx$

$$= \int_0^3 \int_0^1 \left(x^2 z + y\frac{z^2}{2}\right)_{z=-1}^{z=1} dy\, dx$$

$$= \int_0^3 \int_0^1 2x^2\, dy\, dx$$

$$= \int_0^2 2x^2 [y]_0^1\, dx$$

$$= \int_0^3 2x^2 \cdot 1 dx$$

$$= 2\left[\frac{x^3}{3}\right]_0^3$$

$$= 18$$

Exercise Set 8.1

Evaluate the iterated integral.

1. $\int_0^2 \int_0^{z^2} \int_0^{y-z} (2x - y)\, dx\, dy\, dz$

2. $\int_0^1 \int_y^{2y} \int_0^{x+y} 6xy\, dz\, dx\, dy$

3. $\int_1^2 \int_0^{2z} \int_0^{\ln x} x e^{-y}\, dy\, dx\, dz$

4. $\int_0^1 \int_0^1 \int_0^{\sqrt{1-z^2}} \frac{z}{y+1}\, dx\, dz\, dy$

5. $\int_0^{\pi} \int_0^1 \int_0^{\sqrt{1-z^2}} z \sin x\, dy\, dz\, dx.$

[1-5] (Ref: Calculus – James Stewart- 8th Ed. Page:1037)

6. $\int_{-1}^1 \int_0^2 \int_0^1 \left(x^2 + y^2 + z^2\right) dx\, dy\, dz$

7. $\int_{1/3}^{1/2} \int_0^{\pi} \int_0^1 z\, x \sin x\, y\ dx\, dy\, dz$

8. $\int_1^2 \int_{-1}^{y^2} \int_{-1}^{z} y\, z\ dx\, dz\, dy$

9. $\int_0^{\pi/4} \int_0^1 \int_0^{x^2} x \cos y\ dz\, dx\, dy$

10. $\int_0^3 \int_0^{\sqrt{9-z^2}} \int_0^x xy\, dy\, dx\, dz$

11. $\int_1^3 \int_x^{x^2} \int_0^{\ln z} x e^y\ dy\, dz\, dx$

12. $\int_0^2 \int_0^{\sqrt{4-x^2}} \int_{-5+x^2+y^2}^{3-x^2-y^2} x\ dz\, dy\, dx$

[6- 12] (Ref: Calculus – Howard Anton- 10th Ed. Page: 1045)

Application of Triple Integrals

The volume of a closed bounded region R in space is $\iiint_R dV$

Example 2: Find the volume V of the tetrahedron bounded by the planes $x = 0,\ y = 0,\ z = 0$ and $x + y + z = 4$. **(see diagram)**

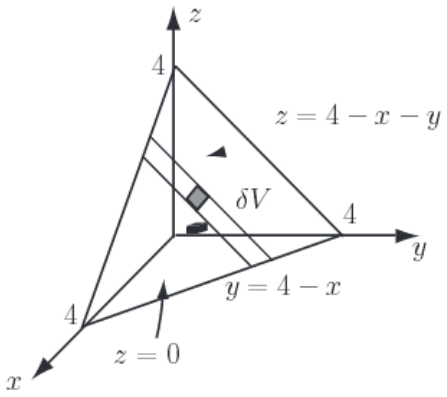
Solution:

$$V = \int_0^4 \int_0^{4-x} \int_0^{4-x-y} dz\, dy\, dx$$

$$= \int_0^4 \int_0^{4-x} (4 - x - y)\, dy\, dx$$

$$= \int_0^4 \left[(4-x)y - \frac{y^2}{2}\right]_0^{4-x} dx$$

$$= \frac{1}{2}\left[\frac{(4-x)^3}{-3}\right]_0^4 = \frac{32}{3}$$



Exercise Set 8.2

Use a triple integral to find the volume of the given solid.

1. The tetrahedron enclosed by the coordinate planes and the plane $2x + y + z = 4$.
2. The tetrahedron enclosed by the coordinate planes and the plane $2x + y + z = 2$.
3. Find the volume V of the tetrahedron bounded by the planes $x = 0,\ y = 0,\ z = 0\ \ x + y + z = 3$.

[1-4] (Ref: Calculus – James Stewart- 8th Ed. Page:1037)

Chapter 9
Ordinary Differential Equations

Differential Equation: A differential equation (DE) is an equation that involves differentials or derivatives. An equation involving a function of one independent variable and its (ordinary) derivatives is called an **ordinary differential equation** (ODE). For example, the equations

$$\frac{dy}{dx} + 2y = 5 \quad \text{and} \quad \frac{d^2y}{dx^2} + y\frac{dy}{dx} + y = e^x \sin x$$

are ordinary differential equations as it contain only ordinary derivatives.

Initial Value Problem (IVP): If a differential equation is required to satisfy conditions on the dependent variable and its derivatives specified at one value of the independent variable, these conditions are called initial conditions, and the problem is called an initial value problem (IVP).

Example: A stone is dropped from the top of a tower of height *h* under gravity can be expressed as,

$\frac{d^2y}{dt^2} = g$ (neglecting friction) with initial conditions when $t = 0,\ y = h, \frac{dy}{dt} = 0$

First Order Differential Equations

An ordinary differential equation of order one can be written as:

$$\frac{dy}{dx} = f(x, y)$$

- **Separation of Variables:** If it is possible to rearrange the terms of the DE in two groups each containing only one variable, the equation is said to be separable. A separable equation is of the form

$$\frac{dy}{dx} = f(x)g(y)$$

Example 1: Solve the following differential equation,

$$xy\frac{dy}{dx} = (1 + x^2)(1 + y^2)$$

Solution: The equation is separable and can be written as, $\frac{y}{1+y^2}dy = \frac{(1+x^2)}{x}dx = \left(\frac{1}{x} + x\right)dx$

Integrating, we get $\frac{1}{2}\ln(1 + y^2) = \ln|x| + \frac{x^2}{2} + \frac{C}{2}$, C is a constant

Simplifying,

$$\ln(1 + y^2) = \ln(x^2) + x^2 + C$$

Example 2: Solve the following IVP: $\frac{dy}{dt} = e^{-y}(2t - 4),\ y(0) = 0$

Solution: The equation is separable and can be written as

$$e^y dy = (2t - 4)dt$$

Integrating, we get $e^y = t^2 - 4t + c$

Using initial condition, we have $c = 1$

Therefore, $e^y = t^2 - 4t + 1 \quad \Rightarrow y(t) = \ln(t^2 - 4t + 1)$

Example 3: Solve the following initial value problem: $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}, \quad y(0) = 1$

Solution: The equation is separable and can be written as $\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$

Integrating, we get $\tan^{-1} y = \tan^{-1} x + c$

Using initial condition, we have $c = \frac{\pi}{4}$. Therefore, $\tan^{-1} y = \tan^{-1} x + \frac{\pi}{4}$.

Exercise Set 9.1

I. Find the solution of the following DEs using separation of variables

1. $\frac{dy}{dt} = (ty^2)$
2. $\frac{dy}{dt} = t^4 y$
3. $\frac{dy}{dt} = 2y + 1$
4. $\frac{dy}{dt} = 2 - y$
5. $\frac{dy}{dt} = e^{-y}$
6. $\frac{dx}{dt} = 1 + x^2$
7. $\frac{dy}{dt} = 2ty^2 + 3y^2$
8. $\frac{dy}{dt} = \frac{t}{y}$
9. $\frac{dy}{dt} = y(1 - y)$
10. $\frac{dy}{dt} = t\sqrt[3]{y}$
11. $\frac{dy}{dt} = \frac{1}{2y+1}$
12. $\frac{dy}{dt} = \frac{2y+1}{t}$
13. $\frac{dy}{dt} = \frac{t}{t^2y+y}$
14. $\frac{dy}{dt} = \frac{4t}{1+3y^2}$
15. $\frac{dy}{dt} = y^2 - 4$
16. $\frac{dw}{dt} = \frac{w}{t}$
17. $\frac{dy}{dt} = \sec y$
18. $\frac{dy}{dt} = \frac{1}{ty+t+y+1}$
19. $\frac{dy}{dt} = \frac{e^t y}{1+y^2}$

[1-19] [Ref: Differential Equations – P. Blanchard, R. L. Devaney, G. R. Hall 4th edition. Page #34]

II. Solve the initial value problems using separation of variables:

1. $\frac{dx}{dt} = -xt, x(0) = \frac{1}{\sqrt{\pi}}$
2. $\frac{dy}{dt} = -y^2, y(0) = \frac{1}{2}$
3. $\frac{dy}{dt} = t^2y^3, y(0) = -1$
4. $\frac{dy}{dt} = \frac{t}{y-t^2y}, y(0) = 4$
5. $\frac{dy}{dt} = ty^2 + 2y^2, y(0) = 1$
6. $\frac{dy}{dt} = \frac{1-y^2}{y}, y(0) = -2$
7. $\frac{dy}{dt} = (y^2 + 1)t, y(0) = 1$
8. $\frac{dy}{dt} = \frac{1}{2y+3}, y(0) = 1$
9. $\frac{dy}{dt} = 2ty^2 + 3y^2t^2, y(1) = -1$
10. $\frac{dy}{dt} = \frac{y^2+5}{y}, y(0) = -2$

[1-10] [Ref: Differential Equations – P. Blanchard, R. L. Devaney, G. R. Hall 4th edition. Page #34]

- **First Order Linear Differential Equations (General Form):** $\frac{dy}{dt} + p(t)y = q(t)$

- First find the integrating factor (IF) by using the following formula, IF $= \mu(t) = e^{\int p(t)dt}$
- Multiply eq (1) by $\mu(t)$.
- Eq (1) can be written as: $\frac{d}{dt}\big(y.\mu(t)\big) = q(t).\mu(t)$
- By integrating, we get the general solution as: $y.\mu(t) = \int [q(t).\mu(t)]dt + c$

Example 4: Find the general solution of

$$y' + y\tan x = \sin 2x$$

Solution: Here $p(x) = \tan x$. An integrating factor is, $e^{\int \tan x\, dx} = e^{\ln \sec x} = \sec x$.

Multiplying by $\sec x$, we have, $\frac{d}{dx}(y.\sec x) = (2\sin x \cos x)\sec x = 2\sin x$

By integrating, $y\sec x = -2\cos x + A$

Or $y = -2\cos^2 x + A\cos x$

Example 5: Find the solution of the IVP $\frac{dy}{dt} + \frac{y}{t} = e^t,\ y(0) = 0.$

Solution: Now, $\mu(t) = e^{\int \frac{1}{t} dt} = e^{\ln t} = t$

Therefore $\frac{d}{dt}(yt) = t\, e^t$

Integration gives, $yt = \int t\, e^t dt = \mathrm{yt} = t\, e^t - e^t + c$

Using initial condition, we have $c = 1$

Therefore, the solution is $yt = t\, e^t - e^t + 1.$

Exercise Set 9.2

I. Find the solution to the following first-order linear differential equations:

1. $\frac{dy}{dt} = -\frac{y}{t} + 2$
2. $\frac{dy}{dt} = \frac{3}{t}y + 2$
3. $\frac{dy}{dt} = -\frac{y}{1+t} + t^2$
4. $\frac{dy}{dt} = -2ty + 4e^{-t^2}$
5. $\frac{dy}{dt} - \frac{2t}{1+t^2}y = 3$
6. $\frac{dy}{dt} - \frac{2}{t}y = t^3e^t$
7. $\frac{dy}{dt} - \frac{2y}{t} = 2t^2$

II. Find the solution to the following Initial Value Problems:

8. $\frac{dy}{dt} = -\frac{y}{1+t} + 2, y(0) = 3$
9. $\frac{dy}{dt} = \frac{y}{1+t} + 4t^2 + 4t, y(1) = 10$
10. $\frac{dy}{dt} - \frac{2y}{t} = 2t^2, y(-2) = 4$
11. $\frac{dy}{dt} = -2ty + 4e^{-t^2}, y(0) = 3$
12. $\frac{dy}{dt} - \frac{2t}{1+t^2}y = 3$
13. $\frac{dy}{dt} - \frac{2}{t}y = t^3e^t$
14. $\frac{dy}{dt} - \frac{3y}{t} = 2t^3e^{2t}, y(1) = 0$

[1-14] [Ref: Differential Equations – P. Blanchard, R. L. Devaney, G. R. Hall 4th edition. Page #133

Exercise Set 9.3

Application of first order linear differential equation.

1. The population of a country will double within 30 years. How many years will it take to be triple? Let the increase rate be proportional to the remaining population.

2. If the growth rate of the number of bacteria at any time t is proportional to the number present at t and doubles in 1 week, how many bacteria can be expected after 2 weeks? After 4 weeks?
3. If the population of a country doubles in 40 years, in how many years will it be four times under the present population, assuming that the rate of increase is proportional to the present population.
4. The population of a town grows at a rate proportional to the population present at time t. The initial population of 500 increases by 15% in 10 years. What will be the population in 30 years? How fast is the population growing at $t = 30$?
5. The radioactive isotope of lead, Pb-209, decays at a rate proportional to the amount present at time t and has a half-life t and has a half-life t of 3.3 hours. If 1 gram of this isotope is present initially, how long will it take for 90% of the lead to decay?
6. The population of bacteria in a culture grows at a rate proportional to the number of bacteria present at time t. After 3 hours it is observed that 400 bacteria are present. After 10 hours 2000 bacteria are present. What was the initial number of bacteria?
7. Newton's law of cooling states that the rate of change of temperature of a body is proportional to the difference between the temperature of the body and that of the surrounding medium. The room temperature is 70^0 F. When a cake is removed from an oven, its temperature is measured at 300° F. Three minutes later, its temperature is$200°$F. How long will it take for the cake to cool to a temperature of $75°$F?
8. A small metal bar, whose initial temperature was 20° C, is dropped into a large container of boiling water. How long will it take the bar to reach 90° C if it is known that its temperature increases 2° in 1 second? How long will it take the bar to reach 98° C?

[1-8] (Ref: Differential Equations *with Modeling Applications* – Dennis G. Zill 11th Ed. Pages: 92,93)

Chapter 10
Higher Order Linear Differential Equations

The general *n*th order linear differential equations with constant coefficients is

$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_{n-1} \frac{dy}{dx} + a_n y = f(x)$,

where $a_0, a_1, \cdots, a_n$ are constants.

In terms of differential operator $\equiv \frac{d}{dx}$, it can be written as

$$(a_0 D^n + a_1 D^{n-1} + \cdots + a_{n-1} D + a_n) y = f(x)$$

Or, in symbolic form, as $L(D)y = f(x)$

Where, $L(D) \equiv a_0 D^n + a_1 D^{n-1} + \cdots + a_{n-1} D + a_n$ is a polynomial in *D* of degree *n*.

- The equation is said to be homogeneous when $f(x) = 0$ and to be non-homogeneous when $f(x) \neq 0$.
- We shall now concentrate on the solution of second order LDEs. The general form of the equation is

$$a_0 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_2 y = 0$$

Where a_0, a_1, a_2 are constants.

A trial solution $y = e^{mx}$ leads to the auxiliary equation (AE)

$$a_0 m^2 + a_1 m + a_2 = 0.$$

Suppose the roots of the auxiliary equation are m_1 and m_2.

The general solution, depending on the nature of the roots, may be expressed in any one of the following forms:

(a) If $m_1 \neq m_2$ and both are real, the solution is, $y = Ae^{m_1 x} + Be^{m_2 x}$.

(b) If $m_1 = m_2 = \alpha$ (say), The two solutions are not independent and become,

$$y = (A + Bx)e^{\alpha x}$$

(c) If m_1 and m_2 are complex, say $\alpha \pm i\beta$, the corresponding general solution is

$$y = e^{\alpha x}(A \cos \beta x + B \sin \beta x)$$

The above rules may be extended to higher-order equations.

Let we are given the auxiliary roots of a certain homogeneous DES, write down the general solution based on the previous discussions,

- If $m = 2, -3, 5$ then $y = c_1 e^{2x} + c_2 e^{-3x} + c_3 e^{5x}$
- If $m = 3, 3$ then $y = (c_1 + c_2 x)e^{3x}$
- If $m = 2 \pm 3i$ then $y = e^{2x}(A \cos 3x + B \sin 3x)$

Example 1: Find the general solution to,

$\frac{d^2 y}{dt^2} + 3\frac{dy}{dt} + 2y = 0.$

Solution: Let, $D \equiv \frac{d}{dt}$, then the given equation can be written as

$$(D^2 + 3D + 2)y = 0 \ldots. (1)$$

Let $y = e^{mt}$ is a trial solution of eqn (1).

Therefore, the auxiliary equation is

$$m^2 + 3m + 2 = 0$$

Solving, $m = -1, -2$ (the roots are real & unequal)

Thus, the general solution is

$$y = c_1 e^{-t} + c_2 e^{-2t}$$

Example 2: Find the general solution to,

$\frac{d^2 y}{dx^2} + 6\frac{dy}{dx} + 9y = 0.$

Solution: Let, $D \equiv \frac{d}{dx}$, then the given equation can be written as

$$(D^2 + 6D + 9)y = 0 \ldots.(1)$$

Let $y = e^{mx}$ is a trial solution of eqn (1).

Therefore, the auxiliary equation is

$$m^2 + 6m + 9 = 0$$

Solving, $m = -3, -3$ (the roots are real & equal)

Thus, the general solution is

$$y = (c_1 + c_2 x)e^{-3x}$$

Example 3: Find the solution of the initial value problem

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 2y = 0,\ y(0) = 0,\ y'(0) = 0.$$

Solution: Let, $D \equiv \frac{d}{dt}$, then the given equation can be written as

$$(D^2 + 2D + 2)y = 0 \ldots. (1)$$

Let, $y = e^{mt}$ is a trial solution of eq. (1).

Therefore the auxiliary equation is, $m^2 + 2m + 2 = 0$

Solving, $m = -1 \pm i$ (the roots are complex) [using calculator or formula]

Thus the general solution is [with α= - 1, β = 1]

$$y = e^{-t}(A\cos t + B\sin t)$$

Now we are going to apply the initial conditions.

$$y = e^{-t}(A\cos t + B\sin t) \ldots\ldots (2)$$

$$y' = -e^{-t}(A\cos t + B\sin t) + e^{-t}(-A\sin t + B\cos t) \ldots\ldots (3)$$

Using, $y(0) = 0$, (2) becomes $0 = 1(A + 0) \Rightarrow A = 0$

Similarly, using, $y'(0) = 0$, (3) gives, $B = 1$, Therefore, $y = e^{-t}\sin t$

Exercise Set 10.1

I. Find the general solution of the following differential equations:

1. $\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 10y = 0.$
2. $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = 0.$
3. $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + y = 0.$
4. $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 7y = 0.$

II. Find the general solution of the second-order differential equation given.

5. $\frac{d^2y}{dt^2} - 6\frac{dy}{dt} - 7y = 0$
6. $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 9y = 0$
7. $\frac{d^2y}{dt^2} + 8\frac{dy}{dt} + 25y = 0$
8. $\frac{d^2y}{dt^2} - \frac{dy}{dt} - 12y = 0$
9. $\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 4y = 0$
10. $\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 29y = 0$

III. Find the general solution of the second-order IVP given

11. $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} - 3y = 0$
    $y(0) = 6,\ y'(0) = -2$
12. $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} - 5y = 0$
    $y(0) = 11,\ y'(0) = -7$
13. $\frac{d^2y}{dt^2} - 8\frac{dy}{dt} + 16y = 0$
    $y(0) = 3,\ y'(0) = 11$
14. $\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 4y = 0$
    $y(0) = 1,\ y'(0) = 1$

15. Consider any damped harmonic oscillator equation, $m\frac{d^2y}{dt^2} + b\frac{dy}{dt} + ky = 0.$

    Solve the initial value problem which is based on the previous equation

    (a) $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 0,\ y(0) = 2,\ v(0) = -3$

    (b) $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 0,\ y(0) = 3,\ v(0) = -5$

[1-15] (Ref: Differential Equations – P. Blanchard, R. L. Devaney, G. R. Hall 4th Ed. Pages: 187, 188, 342)

Non-homogeneous LDEs: The solution of the non-homogeneous linear differential equation, $L(D)y = f(x)$ is of the form, $y = y_c + y_p$, where y_c is the general solution of $L(D)y = 0$ and y_p is a particular solution of $L(D)y = f(x)$.

The particular solution y_p is called a particular integral (PI) and the general solution y_c of the related homogeneous equation is the **complementary function** (CF).

The method of undetermined Coefficient

	Corresponding RHS $f(x)$	**Assumed form of y_p**
1	polynomial of degree n ($n = 0,1,2,3\cdots$)	polynomial of degree n
2	$e^{\alpha x}, \alpha \neq 0$ (exponential)	$Ae^{\alpha x}$
3	$\sin \alpha x$ / $\cos \alpha x$, $\alpha \neq 0$(sine/cosine)	$A \sin \alpha x + B \cos \alpha x$

Note: If the trial PS solutions duplicate terms found in y_c, then multiply the trial solution by x repeatedly until it doesn't. The final trial PS solution is the modified expression.

Example 4: Find the general solution of $\frac{d^2y}{dt^2} + 4y = 3t + 2$

Solution: Let, $D \equiv \frac{d}{dt}$, then the given equation can be written as

$$(D^2 + 4)y = 3t + 2 \quad \ldots\ldots(1)$$

Consider, $(D^2 + 4)y = 0$…….(2)
Let $y = e^{mt}$ is a trial solution of eqn (2).
Therefore the auxiliary equation is $m^2 + 4 = 0, \Rightarrow \mathrm{m} = \pm 2i$
The complimentary function is $y_c = \mathrm{A}\cos 2t + B \sin 2t$
Again let $y_p = a_0 + a_1 t$

$$Dy_p = a_1$$
$$D^2 y_p = 0$$

Substituting the values of y_p and its derivative into the given differential equation (1), we have
$4\,a_0 + 4\,a_1 t = 2 + 3t$
Now, equating like terms, we have,

$$4\,a_0 = 2; 4\,a_1 = 3$$

Solving, $a_0 = \frac{1}{2}$ and $a_1 = \frac{3}{4}$
The general solution is, $y = y_c + y_p = \mathrm{A}\cos 2t + B \sin 2t + \frac{1}{2} + \frac{3}{4}\,t$.

Example 5: Find the general solution of $\frac{d^2y}{dt^2} - \frac{dy}{dt} - 6y = e^{4t}$

Solution: Let, $D \equiv \frac{d}{dt}$, then the given equation can be written as

$$(D^2 - D - 6)y = e^{4t} \ldots\ldots(1)$$

Consider, $(D^2 - D - 6)y = 0$…….(2)
Let, $y = e^{mt}$ is a trial solution of eqn (2).
Therefore, the auxiliary equation is $m^2 - m - 6 = 0$
Solving, $m = -2,\ 3$
The complimentary function is, $y_c = c_1 e^{-2t} + c_2 e^{3t}$

Let, $y_p = Ae^{4t}$
$Dy_p = 4Ae^{4t}$
$D^2 y_p = 16Ae^{4t}$
Eqn (1) implies, $(D^2 - D - 6)y_p = e^{4t}$
$\Rightarrow 16Ae^{4t} - 4Ae^{4t} - 6Ae^{4t} = e^{4t} \Rightarrow 6Ae^{4t} = e^{4t}$
Comparing $6A = 1 \Rightarrow A = \frac{1}{6}$
Therefore $y = c_1 e^{-2t} + c_2 e^{3t} + \frac{1}{6}\,\mathrm{e}^{4t}$

Exercise Set 10.2

I. Compute the general solution with, $y(0) = y'(0) = 0$.

1. $\frac{d^2y}{dt^2} + 2y = 3t + 2$
2. $\frac{d^2y}{dt^2} + 4y = t - \frac{t^2}{20}$

II. Find the general solution of the given differential equation.

3. $\frac{d^2y}{dt^2} - \frac{dy}{dt} - 6y = e^{4t}$
4. $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = 2\,e^{-3t}$
5. $\frac{d^2y}{dt^2} - \frac{dy}{dt} - 2y = 5\,e^{3t}$
6. $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 13y = e^{-t}$
7. $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 13y = -3\,e^{-2t}$
8. $\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 10y = e^{-2t}$
9. $\frac{d^2y}{dt^2} - 5\frac{dy}{dt} + 4y = e^{4t}$
10. $\frac{d^2y}{dt^2} + \frac{dy}{dt} - 6y = 4\,e^{-3t}$

III. Find the general solution of the given initial value problem.

11. $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = e^{-t}$, $y(0) = y'(0) = 0$
12. $\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 12y = 3e^{-t}$, $y(0) = 2,\ y'(0) = 1$
13. $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 13y = -3e^{-2t}$, $y(0) = y'(0) = 0$
14. $\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 10y = e^{-2t}$, $y(0) = y'(0) = 0$

IV. Find the general solution of the mixed problem. Compute the solution with, $y(0) = y'(0) = 0$.

15. $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = e^{-t} + 4$
16. $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = e^{-t} - 4$
17. $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = 2t + e^{-t}$
18. $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = 2t + e^{t}$
19. $\frac{d^2y}{dt^2} + 4y = t + e^{-t}$
20. $\frac{d^2y}{dt^2} + 4y = t + e^{-t}$

[1-20] (Ref: Differential Equations – P. Blanchard, R. L. Devaney, G. R. Hall 4th edition. Pages: 399, 400, 402)

Example 6: Find the general solution of $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = \cos t$

Solution: Let, $D \equiv \frac{d}{dt}$, then the given equation can be written as

$$(D^2 + 3D + 2)y = \cos t \dots\dots (1)$$

Consider, $(D^2 + 3D + 2)y = 0$…….(2)
Let $y = e^{mt}$ is a trial solution of eqn (2).
Therefore the auxiliary equation is $m^2 + 3m + 2 = 0$
Solving $m = -2,\ -1$
The complimentary function is $y_c = c_1e^{-2t} + c_2e^{-t}$
Let, $y_p = A\cos t + B\sin t$

$$Dy_p = -A\sin t + B\cos t$$

$$D^2y_p = -A\cos t - B\sin t$$

Eqn (1) implies, $-A\cos t - B\sin t + 3(-A\sin t + B\cos t) + 2(A\cos t + B\sin t) = \cos t$
Simplifying and equating both sides

$$A + 3B = 1$$
$$-3A + B = 0$$

Solving $A = \frac{1}{10},\ B = \frac{3}{10}$
Therefore the general solution is

$$y = y_c + y_p = c_1e^{-2t} + c_2e^{-t} + \frac{1}{10}\cos t + \frac{3}{10}\sin t$$

Example 7: Solve $\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 10y = e^{-2t}$

Solution: Let, $D \equiv \frac{d}{dt}$, then the given equation can be written as

$$(D^2 + 7D + 10)y = e^{-2t} \dots\dots (1)$$

Consider,$(D^2 + 7D + 10)y = 0$…….(2)
Let $y = e^{mt}$ is a trial solution of eqn(2).
Therefore the auxiliary equation is $m^2 + 7m + 10 = 0$
Solving $m = -5,\ -2$
The complimentary function is $y_c = c_1e^{-5t} + c_2e^{-2t}$
Let, $y_p = Ae^{-2t}$
Note that some terms of y_c match those of y_p. Multiplication once by t is necessary to eliminate duplicates. Then the final form of the particular solution is, $y_p = A\,t\,e^{-2t}$
Therefore proceeding in similar way, we'll have
$3Ae^{-2t} = e^{-2t} \Rightarrow 3A = 1 \Rightarrow A = \frac{1}{3}$
Therefore the general solution is

$$y = y_c + y_p = c_1e^{-5t} + c_2e^{-2t} + \frac{1}{3}\,t\,e^{-2t}$$

Exercise Set 10.3

I. Find the general solution of the given equation.

1. $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = \cos t$
2. $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 5\cos t$
3. $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = \sin t$
4. $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 2\sin t$
5. $\frac{d^2y}{dt^2} + 9y = \cos t$
6. $\frac{d^2y}{dt^2} + 9y = 5\sin 2t$
7. $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = \cos t$
8. $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = -4\cos 3t$
9. $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = \cos 3t$
10. $\frac{d^2y}{dt^2} + 4y = -\cos\frac{t}{2}$
11. $\frac{d^2y}{dt^2} + 4y = 3\cos 2t$

II. Find the general solution of the given IVP.

12. $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = \cos t,$ $y(0) = y'(0) = 0$
13. $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = 2\cos 3t,$ $y(0) = y'(0) = 0$
14. $\frac{d^2y}{dt^2} + 9y = \cos t,$ $y(0) = y'(0) = 0$
15. $\frac{d^2y}{dt^2} + 4y = \sin 3t,$ $y(0) = 2, y'(0) = 0$

[1-15] (Ref: Differential Equations – P. Blanchard, R. L. Devaney, G. R. Hall 4th Ed. Pages: 412, 424)

Chapter 11

System of Linear Differential Equations

Example 1: Find a general solution of the following system:

$\frac{dx}{dt} = 2x - 2y$ …..(1)

$\frac{dy}{dt} = 2x + 2y$ …..(2)

Solution: Using a differential operator $D = \frac{d}{dt}$, we can write $Dx = 2x - 2y$

$Dy = 2x + 2y$

This system can be written as:

$(D-2)x + 2y = 0$ ……(3)

$2x - (D-2)y = 0$ ……(4)

Multiplying eqn. (3) by 2 and operating on (4) with $(D-2)$ we get,

$2(D-2)x + 4y = 0$ ….(5)

$2(D-2)x - (D^2 - 4D + 4)y = 0$ …..(6)

Subtracting (6) from (5), $(D^2 - 4D + 8)y = 0$

Auxiliary eqn. is $m^2 - 4m + 8 = 0$

and its solutions are $m = 2 \pm 2i$

Thus $y(t) = e^{2t}(A\cos 2t + B\sin 2t)$

Substituting y in (2), we get

$\frac{d}{dt}e^{2t}(A\cos 2t + \mathrm{B}\sin 2t)$

$= 2x + 2e^{2t}(A\cos 2t + B\sin 2t)$

$\Rightarrow e^{2t}(-2A\sin 2t + 2\mathrm{B}\cos 2t) + 2\mathrm{e}^{2\mathrm{t}}(\mathrm{A}\cos 2t + \mathrm{B}\sin 2t) = 2x + 2e^{2t}(A\cos 2t + B\sin 2t)$

$\Rightarrow 2e^{2t}(B\cos 2t - A\sin 2t) = 2x$

$\Rightarrow x(t) = e^{2t}(B\cos 2t - A\sin 2t)$

Solutions are, $x(t) = e^{2t}(B\cos 2t - A\sin 2t)$ and $y(t) = e^{2t}(A\cos 2t + B\sin 2t)$.

Example 2: Solve the following initial value problem:

$\frac{dx}{dt} = -x + y$ ……. (1)

$\frac{dy}{dt} = -3x - 5y$ ….(2) $x(0) = -2$ and $y(0) = 0$

Solution: Using a differential operator $D = \frac{d}{dt}$, we can write $Dx = -x + y$

$Dy = -3x - 5y$

This system can be written as:

$(D+1)x - y = 0$ …..(3)

$3x + (D+5)y = 0$ …..(4)

Multiplying eqn. (3) by $(D+5)$ and adding with (4) gives,$(D^2 + 6D + 8)x = 0$ ……(5)

Auxiliary eqn. is $m^2 + 6m + 8 = 0$

and its solutions are $m = -4, -2$

Therefore, the solution of the given IVP is $x(t) = e^{-4t} - 3e^{-2t}$ & $y(t) = -3e^{-4t} + 3e^{-2t}$

Thus, $x(t) = c_1e^{-4t} + c_2e^{-2t}$…..(6)

Substituting y in (1), we get

$\frac{d}{dt}(c_1e^{-4t} + c_2e^{-2t}) = -(c_1e^{-4t} + c_2e^{-2t}) + y$

$\Rightarrow y(t) = -3c_1e^{-4t} - c_2e^{-2t}$…..(7)

Applying the initial conditions gives,

$c_1 + c_2 = -2$

$-3c_1 - c_2 = 0$

Solving the above system, $c_1 = 1$ and $c_2 - 3$

Exercise Set 11

Solve the following system of linear differential equations.

1. $\frac{dx}{dt} = 2x + 2y$
   $\frac{dy}{dt} = x + 3y$
2. $\frac{dx}{dt} = 2x + 3y$
   $\frac{dy}{dx} = -4y$
3. $\frac{dx}{dt} = x + y$
   $\frac{dy}{dt} = 4x + y$
4. $\frac{dx}{dt} = -x$
   $\frac{dy}{dt} = y - 1$
5. $\frac{dx}{dt} = x + 2y$
   $\frac{dy}{dt} = -y$
6. $\frac{dx}{dt} = 2x$
   $\frac{dy}{dt} = y$
7. $\frac{dy}{dt} = -v$
   $\frac{dv}{dt} = y$
8. $\frac{du}{dt} = u - 1$
   $\frac{dv}{dt} = v - 1$
9. $\frac{dx}{dt} = x$
   $\frac{dy}{dt} = -y$
10. $\frac{dx}{dt} = x$
    $\frac{dy}{dt} = 2y$
11. $\frac{dx}{dt} = x - 1$
    $\frac{dy}{dt} = -y$
12. $\frac{dx}{dt} = x - 2y$
    $\frac{dy}{dt} = -y$

[1-12] (Ref: Differential Equations – P. Blanchard, R. L. Devaney, G. R. Hall 4th edition. Page #178, 180)